Week 6 (LAB3): Block Design

PSTAT122: Design and Analysis of Experiments

Ali Abuzaid

Submission Instructions

- This LAB must be completed and submitted **individually**. Collaboration is allowed for discussion, but each student must submit their own work.
- Ensure that all `R` code are presented clearly and appropriately.
- All figures should be numbered, and axes must be labeled.
- Please use the provided `LAB 3.qmd` file to type your solutions and submit the completed LAB as a PDF file.

You can utilize `RStudio` for this purpose. For guidance, refer to the [Tutorial: Hello, Quarto]).

- – Submit your solution via **Gradescope** as **a single PDF file**.

Due Date

Due Date: Friday, February 13, 2026, 11:59 PM

Overview

In this lab, we explore blocking in experimental design using both a Randomized Complete Block Design (RCBD) and a Balanced Incomplete Block Design (BIBD). You will learn how blocking reduces experimental error and increases the precision of treatment comparisons, and how incomplete blocks can be handled when not all treatments can appear in every block.

1 Objectives:

1- Explain the purpose of blocking in experimental design.

2- Describe the structure and parameters of a Balanced Incomplete Block Design (BIBD).

3- Construct and visualize RCBD and BIBD in `R`.

4- Fit and interpret ANOVA models for both RCBD and BIBD, including blocks.

5- Check ANOVA assumptions using residual plots and normality tests.

6- Interpret treatment effects using pairwise comparisons and Tukey HSD when appropriate.

2 Introduction: Why Blocking?

In many experiments, external sources of variation — such as testers, locations, or days — can influence the results.

To reduce the impact of these unwanted differences, we group similar experimental units together into blocks.

However, sometimes each block cannot test all treatments due to limitations (e.g., time, fatigue, or cost). In such cases, we use a Balanced Incomplete Block Design (BIBD) — where:

Not all treatments appear in every block.

The design is balanced: every pair of treatments appears together in the same number of blocks.

3 Insulation Cut and Strength

> **Insulation Cut and Strength (15 Points)**
>
> The data in `insulate1.txt` is from an experiment that measured impact strength of insulation with the purpose of determining if there is a difference between pieces that are cut crosswise or lengthwise from the material stock lots. The file contains the strength measured in foot-pounds, the type of cut (either cross or lengthwise), and the lot number from which the sample was cut.

Let's start by reading the data

```r
insulate <- read.table("insulate1.txt", header = TRUE)
insulate$Cut <- as.factor(insulate$Cut)
insulate$Lot <- as.factor(insulate$Lot)
head(insulate)
```

```
  Lot    Cut Strength
1   1  Cross     0.46
2   1 Length     1.15
3   2  Cross     0.84
4   2 Length     1.16
5   3  Cross     0.52
6   3 Length     0.81
```

```r
freq_table <- table(insulate$Cut, insulate$Lot )
freq_table
```

```
        1 2 3 4 5
  Cross  1 1 1 1 1
  Length 1 1 1 1 1
```

The three key variables are:

strength: The response variable (dependent variable) measured in foot-pounds. This is the output (yield) we are trying to analyze and explain.

cut_type: The treatment factor (independent variable) of primary interest, with two levels: “cross” and “length”.

lot_number: The blocking factor, a nuisance variable intended to control for variations between different stock lots the yield is strength of insulation

3.1 RCBD Example: Insulation Strength

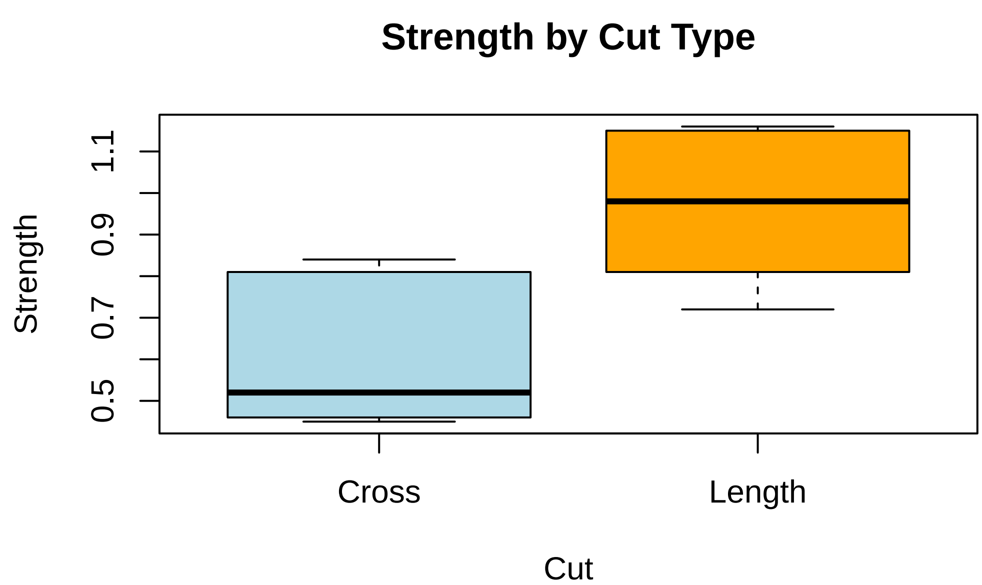
Factor	Levels	Role	Notes
Cut	2	Treatment	Type of cut
Lot	10	Block	Stock lots

! Question

a) Draw a boxplot and perform a one-way ANOVA to determine whether or not the type of cut effects the impact strength. What would we conclude from this simple model?

- The purpose of Blocking in Experimental Design Blocking is used to increase the accuracy of an experiment by isolating nusuance variables. Nusaince variables are factors that are known to influence the response variable like Lot number in the insulation study of Taster in the coffee study. This reduces the "noise" error in the data making it easier to see if the actual treatment is making a statistically significant difference. *Structure and Parameters of Balenced Incomplete Block Design Used when a blaock is too small to contain all treatments. It is defined by Ti, the total number of treatments being studies, b the total number of blocks, r replications or how many times each treatment appears across the whole experiment, k block size meaning how many treatments are in a single bloack and Lambda which is the number of times every pair of treatments appears together in the same block esuring the design is balanced.

```r
boxplot(Strength ~ Cut, data = insulate,
        main = "Strength by Cut Type",
        col = c("lightblue", "orange"))
```



```r
model_simple <- aov(Strength ~ Cut, data = insulate)
summary(model_simple)
```

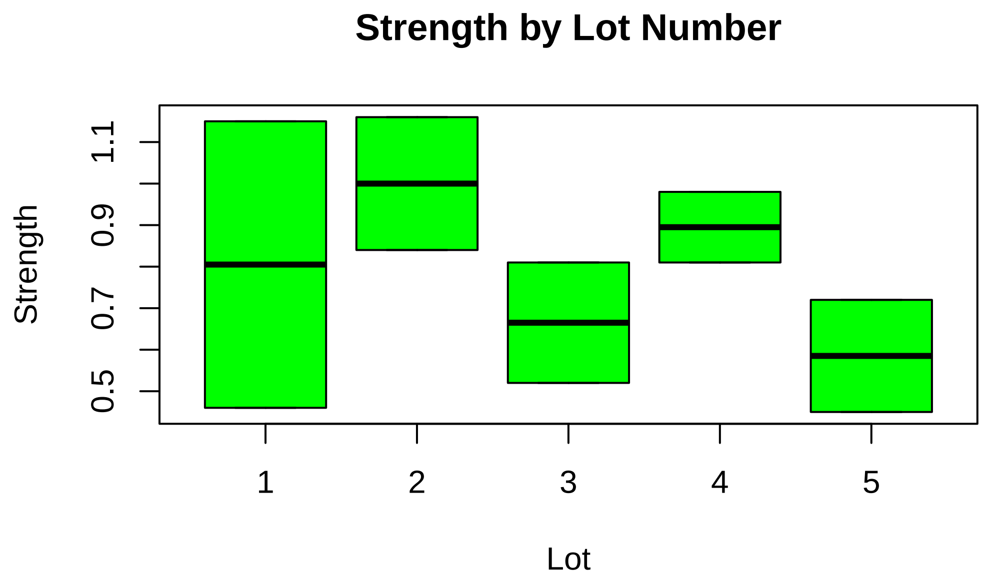
```
            Df Sum Sq Mean Sq F value Pr(>F)
Cut          1 0.3028 0.30276    7.93 0.0226 *
Residuals    8 0.3054 0.03818
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

-The boxplot shows a clear difference in median impact strength between the two groups. The length cut generally results in higher impact strength that the Cross cut. -The p-value for summary of model simple THE P_VALUE(0.0226) $<$ 0.05 SO WE CONCLUDE that the type of cut has a statistically significant effect on impact strength. -While the model may show a significant effect, the simple model is likely ineffecient because we are ignoring Lot number and the residuals is very large. This makes the test fro Cut have less power than it would be if we accounted for variability.

The boxplot shows a difference between the type of cut. The analysis of variance is conducted below:

> **❗ Question**
>
> b) Draw a boxplot of the strengths by lot number this time. Do you think it will make a difference if we use this additional factor in the analysis?

```r
boxplot(Strength ~ Lot, data = insulate,
        main = "Strength by Lot Number",
        col = "green")
```



-The media strength levels vary significantly from eachother. - Yes using this additional factor in the analysis will make a significant difference because the nusiance variation is confirmed by the boxplot to be the material stock lots being different. This difference in material stock lots it increasing the residual sum of squares in the simple anova model. This takes away the lot to lot variability from the error term and increases model power. It also decreases the MSE and makes it easier to detect if the Cut type is actually significant.

> **!** Question
>
> c) Perform an ANVOA testing the same question as in part (a) but using the lot number as a blocking variable. How does this affect our conclusion?

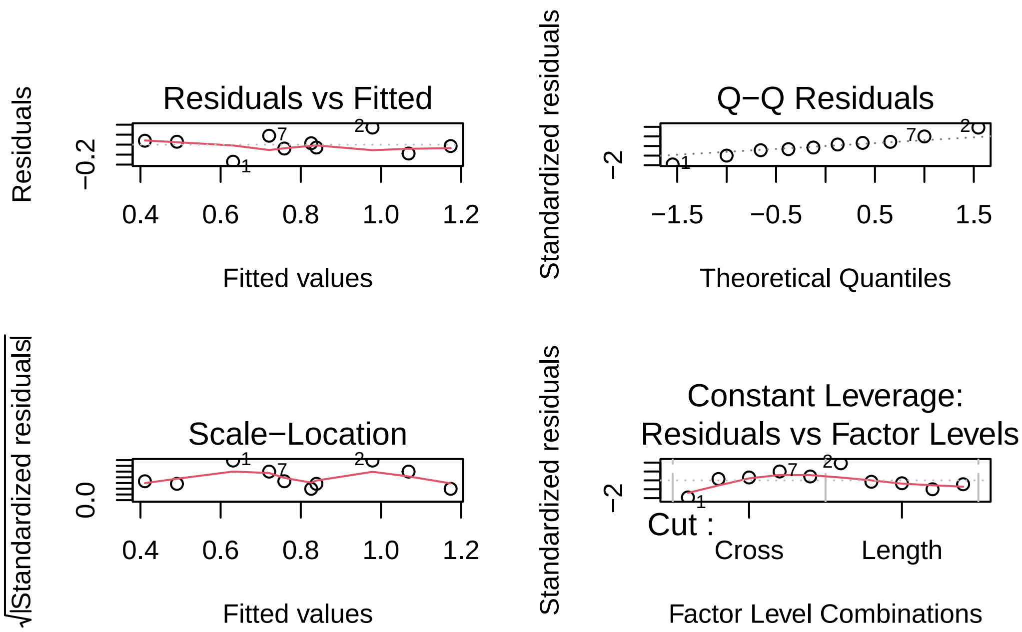
```r
model_rcbd <- aov(Strength ~ Cut + Lot, data = insulate)
summary(model_rcbd)
```

```
            Df  Sum Sq Mean Sq F value Pr(>F)  
Cut          1 0.30276 0.30276  15.245 0.0175 *
Lot          4 0.22600 0.05650   2.845 0.1677  
Residuals    4 0.07944 0.01986                 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Using the lot number as a blocking variable reduces the unexplained experimental error (residual sum of squares). In this model the p-value for Cut decreases to 0.0175 leading to a more accurate F-test and provides a higher condidence that the cut type impacts strength after we controll for lot differences.

> **!** Question
>
> d) check the validity of the model.

```r
par(mfrow = c(2, 2))
plot(model_rcbd)
```




```r
shapiro.test(residuals(model_rcbd))
```

```
	Shapiro-Wilk normality test

data:  residuals(model_rcbd)
W = 0.98501, p-value = 0.9863
```

Constant variance - in the residuals vs fitted plot the points are RANDOMLY SCATTERED WITHOUT ANY CLEAR PATTERN. THE SPREAD IS RELATIVELY EVEN, WE ASSUME THE CONSTANT VARIANCE IS MET

Normality - In the QQ plot THE POINTS STAY CLOSE TO THE DASHED DIAGONAL LINE. THE ASSUMPTION THAT THE RESIDUALS ARE NORMALLY DISTRIVUTED IS MET AND ANOVA RESULTS ARE VALID

Scale_Location - checks homoscedasticity. The red line is relatively horizontal and the spread of points does not not significantly change as the fitted values increase. This confirms the variance is consistent

Residuals vs Leverage - checks for influential observations / outliers. All data points fall within Cooks distance boundaries meaning there are no highly influential outliers that are skweing my results.

Shapiro: -THE P VALUE FOR THE SHAPIRO TEST is greater than 0.05 (.9863) so we fail to reject the null hypothesis. This strengthens the conclusion that the residuals are normally distributed, the value is super high which is great

4 Coffee Taste Study (15 Points)

Imagine you are a sensory analyst testing 6 new coffee blends (A–F).

You have 10 tasters available, but each can only taste 3 blends (to avoid fatigue).

You need a fair design where each blend:

Appears the same number of times across tasters, and

Each pair of blends appears together in the same number of tasters.

This is a perfect setup for a **Balanced Incomplete Block Design.**

4.1 Generate the BIBD Design

```
library("agricolae")

set.seed(100)
treatments <- paste0("Blend", LETTERS[1:6])

# Then pass the variable (not the expression) to design.bib()
design <- design.bib(trt = treatments, k = 3, r = 5, seed = 123)
```

```
Parameters BIB
==============
Lambda     : 2
treatmeans : 6
Block size : 3
Blocks     : 10
Replication: 5

Efficiency factor 0.8

<<< Book >>>
```

```r
head(design$book)
```

```
  plots block treatments
1   101     1     BlendB
2   102     1     BlendF
3   103     1     BlendC
4   201     2     BlendD
5   202     2     BlendC
6   203     2     BlendA
```

Let's explore the used parameters:

$v = 6:$ 6 blends (treatments)

$b = 10:$ 10 tasters (blocks)

$r = 5:$ each blend appears in 5 blocks

$k = 3:$ each block contains 3 blends

$\lambda = 2:$ each pair of blends appears together in 2 blocks

> **❗ Question**
>
> a) Why do we call this an incomplete design? (Hint: does every taster try all blends?)

We call this an incomplete design because the block size k=3 is smaller than the total number of treatments v = 6. In a complete design, like the insulation study above, every taster trys all six coffee blends. Here, every individual taster only trys a subset, 3 out of the 6 of the available blends. Becuase no single taster evaluates every treatment, the blocks are incomplete. The balanced incomplete design ensures that every coffee blend is compared against every other coffee blend an equal number of times lambda = 2, making the analysis fair.

4.2 Simulate Taste Scores

We'll now simulate the actual tasting results. Each taster (block) gives a score (1–10) for each coffee blend they try.

We'll assume Blend A is slightly preferred on average, but there's natural variability among tasters.

```r
library(dplyr)
set.seed(100)
data <- design$book %>% rename(trt = treatments) %>%
  mutate(Score = round(7 +
  ifelse(trt == "BlendA", 1, 0.5) +  # Blend A slightly better
      rnorm(n(), 0, 1), 1))

head(data)
```

```
  plots block    trt Score
1   101     1 BlendB   6.8
2   102     1 BlendF   7.7
3   103     1 BlendC   7.6
4   201     2 BlendD   5.5
5   202     2 BlendC   8.8
6   203     2 BlendA   9.6
```

4.3 Explore the Data

Let's take a look at the overall structure.

```r
str(data)
```

```
'data.frame':	30 obs. of  4 variables:
 $ plots: num  101 102 103 201 202 203 301 302 303 401 ...
 $ block: Factor w/ 10 levels "1","2","3","4",..: 1 1 1 2 2 2 3 3 3 4 ...
 $ trt  : Factor w/ 6 levels "BlendA","BlendB",..: 2 6 3 4 3 1 4 2 1 5 ...
 $ Score: num  6.8 7.7 7.6 5.5 8.8 9.6 7.6 6.1 10.2 6.5 ...
```

```r
summary(data$Score)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  5.500   6.800   7.250   7.447   7.675  10.200 
```

Variables:

`plots` – unique identifier for each observation.

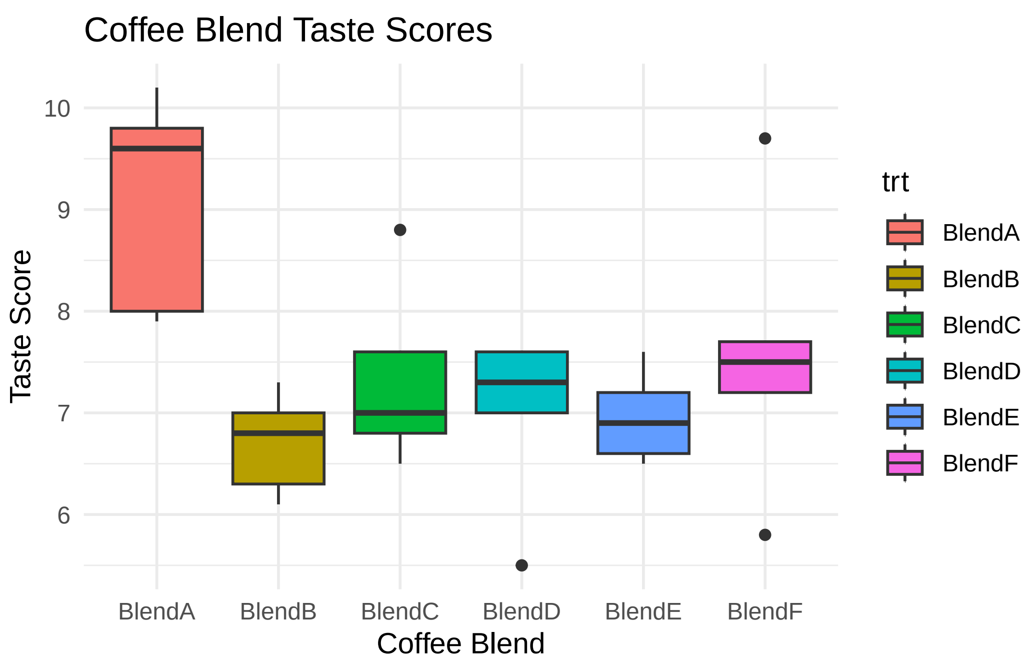
`block` – taster (10 total).

`trt` – coffee blend label (BlendA–BlendF).

`Score` – numeric taste rating.

4.4 Visulization

```r
library(ggplot2)
ggplot(data, aes(x = trt, y = Score, fill = trt)) +
geom_boxplot() +
labs(title = "Coffee Blend Taste Scores",
x = "Coffee Blend",
y = "Taste Score") +
theme_minimal()
```



> **!** Question
>
> b) What does this boxplot suggest about differences between blends? Can you already spot one that looks better on average?

The boxplot clearly shows Blend A looks better on average. Its median line is higher than all other blends and the entire box is shifter upwards on the y-axis meaning it consistently has higher scores. Most of the other blends B-F have overlapping boxes yet Blend B seems to have the lowest median and a wider range of lower scores compared the the other blends. E is rated not too much better than B. The whiskers and height of the boxes show that there is still nautral varaibility within each blends(especially for F) rating likely caused by individual rpreferences of tasters - the blocking factor. Even though Blend A appears better, the anova model is needed to confirm this diffreence is statistically significant. Also, taster bias could affect these scores as some blends might just be tasted by harsher tasters

4.4.1 BIBD Example: Coffee Taste Study

Factor	Levels	Role	Notes
Blend	6	Treatment	Coffee blend
Taster	10	Block	Sensory panel

4.5 Fit the BIBD Model

Now we analyze the data, including both treatment and block effects:

```r
model <- aov(Score ~ trt + block, data = data)
summary(model)
```

```
            Df Sum Sq Mean Sq F value Pr(>F)  
trt          5 18.783   3.757   4.118 0.0149 *
block        9  6.908   0.768   0.841 0.5914  
Residuals   15 13.684   0.912                 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **!** Question
>
> c) Is the treatment effect significant?
>
> Why might including block in the model reduce experimental error?

The treatment effect is significant because the p value for treatment is 0.0149 which is less than alpha = 0.05 making us reject the null hypothesis and conclude there is a statistically significant diffrence in taste scores between the difference coffee blends.

Blocking in the model would reduce experimental error of Taster in the model. It would reduce experimental error by taking out the variation caused by indicidual human preference. It makes the residual MSE smaller and our F test more sensitive and powerful. F changes from 4.378 to 4.118 with blocking. It is a neccesary adjustment because it ensures a fair comparison even though not every taster tried every blend.

Try to fit the ANOVA with only treatment!!

```r
model_no_block <- aov(Score~trt, data=data)
summary(model_no_block)
```

```
            Df Sum Sq Mean Sq F value  Pr(>F)   
trt          5  18.78   3.757   4.378 0.00566 **
Residuals   24  20.59   0.858                   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
library(emmeans)
em_blends <- emmeans(model, specs = ~ trt)
pairs(em_blends)
```

```
 contrast        estimate    SE df t.ratio p.value
 BlendA - BlendB   2.6250 0.675 15   3.887  0.0150
 BlendA - BlendC   2.2167 0.675 15   3.282  0.0470
 BlendA - BlendD   2.7583 0.675 15   4.084  0.0103
 BlendA - BlendE   2.2583 0.675 15   3.344  0.0419
 BlendA - BlendF   1.6917 0.675 15   2.505  0.1837
 BlendB - BlendC  -0.4083 0.675 15  -0.605  0.9890
 BlendB - BlendD   0.1333 0.675 15   0.197  0.9999
 BlendB - BlendE  -0.3667 0.675 15  -0.543  0.9933
 BlendB - BlendF  -0.9333 0.675 15  -1.382  0.7367
 BlendC - BlendD   0.5417 0.675 15   0.802  0.9627
```

```
 BlendC - BlendE   0.0417 0.675 15   0.062  1.0000
 BlendC - BlendF  -0.5250 0.675 15  -0.777  0.9673
 BlendD - BlendE  -0.5000 0.675 15  -0.740  0.9734
 BlendD - BlendF  -1.0667 0.675 15  -1.579  0.6229
 BlendE - BlendF  -0.5667 0.675 15  -0.839  0.9552

Results are averaged over the levels of: block
P value adjustment: tukey method for comparing a family of 6 estimates
```

The pairwise comparison shows the following significant contrasts: Blend A - Blend B (p = 0.0150) Blend A - Blend C (p = 0.0470) Blend A - Blend D (p = 0.0103) Blend A - Blend E (p = 0.0419) Therefore we see that blend A is significantly different from alomst every other blend (except F). Since the estimate values for these contrasts are all positive this means blend A has a significantly higher mean score than Blends B, C, D, and E None of the other blends have a significant difference.

4.6 Compare Treatments

If the treatment effect is significant, we can perform pairwise comparisons

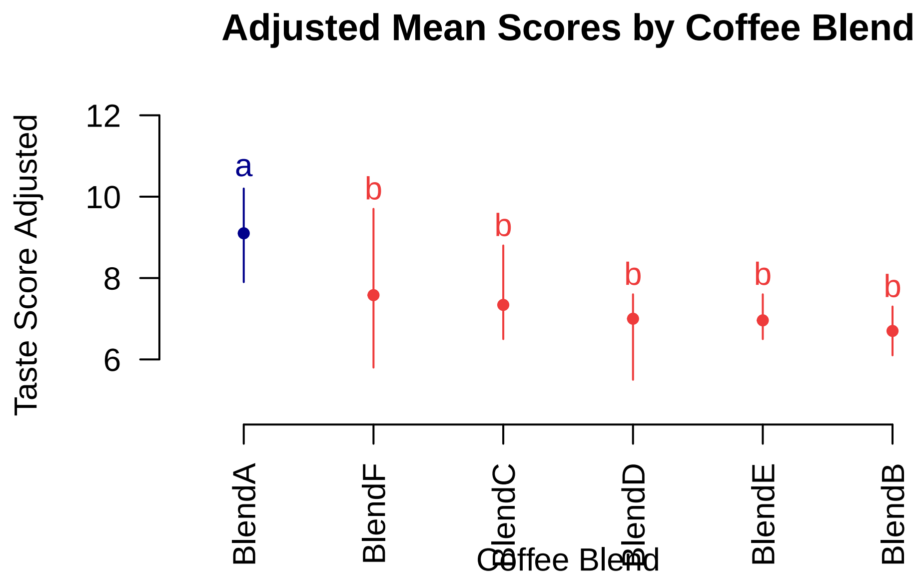
> **!** Question
>
> d) Compare the estimated mean scores (adjusted for blocks) with confidence intervals by using appropriate plot.

```r
library(agricolae)
out <- LSD.test(model, "trt", alpha = 0.05, console = TRUE)
```

```
Study: model ~ "trt"

LSD t Test for Score

Mean Square Error:  0.9122593

trt,  means and individual ( 95 %) CI

       Score       std r        se      LCL       UCL Min  Max Q25 Q50 Q75
BlendA  9.10 1.0723805 5 0.4271438 8.189564 10.010436 7.9 10.2 8.0 9.6 9.8
BlendB  6.70 0.4949747 5 0.4271438 5.789564  7.610436 6.1  7.3 6.3 6.8 7.0
BlendC  7.34 0.9099451 5 0.4271438 6.429564  8.250436 6.5  8.8 6.8 7.0 7.6
```

```
BlendD  7.00 0.8746428 5 0.4271438 6.089564  7.910436 5.5  7.6 7.0 7.3 7.6
BlendE  6.96 0.4505552 5 0.4271438 6.049564  7.870436 6.5  7.6 6.6 6.9 7.2
BlendF  7.58 1.3989282 5 0.4271438 6.669564  8.490436 5.8  9.7 7.2 7.5 7.7

Alpha: 0.05 ; DF Error: 15
Critical Value of t: 2.13145

least Significant Difference: 1.28755

Treatments with the same letter are not significantly different.

       Score groups
BlendA  9.10      a
BlendF  7.58      b
BlendC  7.34      b
BlendD  7.00      b
BlendE  6.96      b
BlendB  6.70      b
```

```r
plot(out,
     main = "Adjusted Mean Scores by Coffee Blend",
     xlab = "Coffee Blend",
     ylab = "Taste Score Adjusted",
     las = 2)
```



Mean Comparison - plot shows Blend A has the highest adjusted mean score around 9 - blends b and e have the lowest medians - blend D has the lowest outlier - groups B C D E F are in group 'b' - group A doesn't share a letter with any other blend meaning it is significantly better than all other tested blends - any blends that do not share a letter are SIGNIFICANTLY DIFFERENT - if the gap between TWO MEANS IS LARGER THANT THE LSD = 1.28755 WE CAN CONCLUDE THOSE TWO COFFEE BLENDS ARE STATISTICALLY DIFFERENT –> GROuP A significantly different